Sahaj Bhadja

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Summary

Product Engineer with 2+ years building **React** and JavaScript applications, **multi-tenant SaaS** platforms, real-time integrations, scalable data models, and AI-powered workflows with strong ownership across testing, performance, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Software Engineering, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant SaaS** platform on Azure SQL and Blob Storage with normalized schemas and row-level isolation, supporting 500+ users while indexed queries reduced dashboard latency from 1.2 seconds to 350 milliseconds.
- Built asynchronous application workflows with explicit job states, validation, and failure isolation, processing 1K+ AI jobs at triple throughput while sustaining 99%+ **reliability** across 5K+ monthly production tasks.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed reusable **React** interfaces and Flask services for uploads, grading history, feedback, and model selection, supporting 150+ instructors and 10K+ daily requests while reducing feedback turnaround 60%.
- Integrated GPT-4o, Claude, and Qwen-VL through configurable model-routing workflows with OCR fallbacks and output validation, applying **LLM capabilities** while maintaining 90%+ grading consistency during production use at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed third-party integration modules using object-oriented patterns, pagination, caching, and field filtering, improving response times 30%, reducing technical debt 20%, and strengthening **application performance** across a cross-platform data-sharing product.
- Automated 20+ **system, unit, and integration tests** with PyTest, Mockito, and Jenkins across Windows, Linux, and macOS, collaborating through Agile reviews and release validation with Microsoft and GitHub stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a configurable resource-management platform using **React and JavaScript**, implementing eight modular services for authentication, resources, billing, leads, permissions, and notifications while synchronizing 2M+ enterprise records across locations.
- Delivered real-time WebSocket data flows and REST integrations through iterative **code reviews**; Jenkins CI/CD, Docker staging, and automated tests achieved 95%+ build validation and reduced deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into an end-to-end architecture with 12+ **integration points**, mapping bookings, pricing, coupons, payments, and status flows before presenting technical solutions and reducing design errors 25%.
- Developed Node.js services and Firebase **real-time data mapping** for 50K+ booking records, maintaining sub-200ms responses across web and mobile clients while documenting configurable workflows for stakeholder review.

Skills

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3
Product Engineering: React, Next.js, Responsive Design, Interface Design, User Experience, Data Structures, Data Mapping, Multi-Tenant Architecture
Backend & Data: Node.js, Express.js, REST APIs, GraphQL, WebSockets, PostgreSQL, Azure SQL, MongoDB, Firebase
AI & Automation: LLMs, Claude, GPT-4o, Prompt Engineering, Workflow Automation, Output Validation
Quality & Cloud: System Testing, Unit Testing, Integration Testing, Code Review, Performance Optimization, AWS, Azure, Docker, Jenkins, CI/CD, Agile/Scrum

Projects

CaseFlow Studio | *React, TypeScript, Node.js, PostgreSQL, WebSockets, Docker*

- Designed a configurable **process automation** application with reusable forms, record views, approval rules, and real-time status flows, mapping enterprise data into versioned workflows with role-based task ownership.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*

- Built a **scalable SaaS application** with secure bank linking, responsive interfaces, MongoDB-backed filtering, and real-time updates across 1M+ transactions, deployed through Kubernetes and automated CI/CD pipelines for enterprise workloads.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Sentry*

- Built configurable appointment and patient-record **workflows** with secure consent-based integrations, responsive dashboards, Twilio notifications, and Sentry monitoring, reducing administrative workload 30% while reliably supporting 100+ daily appointments.